

Diagnostic Teacher Key

Each item identifies the intended evidence, likely misconception, and a concrete next instructional move.

STANDARD	LEARNING INTENTION	SUCCESS EVIDENCE
6.RP.1	Compare two quantities and build the meaning of a ratio.	Name the quantities, preserve order, label the ratio, and identify the whole.

Use: This is not only an answer key. Mark the misconception pattern that best explains each error, then use the listed move during the next warm-up review, table conference, or small-group reteach.

1. Multiplicative comparison

ANSWER

Multiplicative. The scale factor is 4 because $7 \times 4 = 28$.

WHAT THIS TESTS

Whether the student recognizes a factor relationship instead of focusing only on the difference of 21.

LIKELY MISCONCEPTION

Calls the comparison additive because $28 - 7 = 21$, or names 21 as the relationship.

EVIDENCE TO LOOK FOR

Student explicitly states '4 times as many' and connects both quantities with $7 \times 4 = 28$.

NEXT TEACHER MOVE

Build four groups of 7 counters. Ask what repeats and what 21 describes.

2. Additive comparison

ANSWER

Additive. Justin is 12, and the 4 describes the age difference: $8 + 4 = 12$.

WHAT THIS TESTS

Whether the student can distinguish a difference from a multiplicative comparison.

LIKELY MISCONCEPTION

Treats '4 older' as '4 times as old' or says 4 is the ratio.

EVIDENCE TO LOOK FOR

Student uses addition or subtraction and explains that 4 is the amount between the ages.

NEXT TEACHER MOVE

Compare $8 + 4$ with 8×4 on a number line; ask which result matches the situation.

3. Three forms, same order

ANSWER

3 strawberry bars to 5 blueberry bars; 3:5; 3 strawberry bars / 5 blueberry bars.

WHAT THIS TESTS

Ratio order and equivalence across words, colon notation, and labeled fractional form.

LIKELY MISCONCEPTION

Reverses 3 and 5, omits labels, or changes the quantities when changing notation.

EVIDENCE TO LOOK FOR

All three forms keep strawberry first and blueberry second.

NEXT TEACHER MOVE

Have the student underline the first named quantity and circle the second before writing numbers.

4. Find the whole

ANSWER

2 apple bars : 10 total bars; 2 apple bars / 10 total bars; part-to-whole.

WHAT THIS TESTS

Whether the student constructs the whole by adding all three categories.

LIKELY MISCONCEPTION

Uses 2:5 or 2:3 because the student compares apple bars to one other part.

EVIDENCE TO LOOK FOR

Student calculates $5 + 3 + 2 = 10$ and labels 10 as total bars.

NEXT TEACHER MOVE

Put all ten bars inside one drawn boundary and ask what the boundary represents.

5. Part-to-part

ANSWER

5 sports movies : 3 science-fiction movies; 5 sports movies / 3 science-fiction movies; part-to-part.

WHAT THIS TESTS

Whether the student identifies two categories as two parts rather than part and whole.

LIKELY MISCONCEPTION

Calls any ratio written like a fraction part-to-whole or uses 10 as the second quantity.

EVIDENCE TO LOOK FOR

Student keeps 5 and 3, labels both movie categories, and classifies part-to-part.

NEXT TEACHER MOVE

Ask: 'Does either number count every movie on the list?'

6. Part-to-whole

ANSWER

2 comedies : 10 total movies; 2 comedies / 10 total movies; part-to-whole. The whole is all movies on the list.

WHAT THIS TESTS

Whether the student recognizes a named total and preserves the requested order.

LIKELY MISCONCEPTION

Uses 2:5 or 2:3, or reverses the ratio to 10:2.

EVIDENCE TO LOOK FOR

Student totals $5 + 3 + 2$, writes 2 first, and names all ten movies as the whole.

NEXT TEACHER MOVE

Have the student say the ratio aloud before writing: 'comedies to total movies.'

7. Error analysis

ANSWER

The student used the yellow part as the whole. There are 5 counters total, so blue to whole is 3:5. The 3 counts blue counters; the 5 counts every counter.

WHAT THIS TESTS

Direct diagnosis of the most likely part-to-whole misconception.

LIKELY MISCONCEPTION

Changes the label but keeps 3:2, or writes 5:3 because the total is larger.

EVIDENCE TO LOOK FOR

Student identifies why 2 is not the whole and explains both numbers in 3:5.

NEXT TEACHER MOVE

Rebuild the model and ask the student to sweep a finger around all counters before naming the total.

8. Transfer

ANSWER

12 prefer wheat bread. Sample part-to-part: 19 white : 12 wheat. Sample part-to-whole: 19 white : 31 total or 12 wheat : 31 total.

WHAT THIS TESTS

Transfer to a situation where one part is not directly stated and multiple valid ratios are possible.

LIKELY MISCONCEPTION

Uses 31 as the wheat count, subtracts incorrectly, or reverses an unlabeled ratio.

EVIDENCE TO LOOK FOR

Student finds $31 - 19 = 12$, labels each ratio, and explains that reversing order changes the comparison.

NEXT TEACHER MOVE

Draw a bar of 31 split into 19 and an unknown part, then connect each requested phrase to the diagram.

Quick sorting guide

Observed pattern	Instructional response
Difference used as the ratio	Return to equal groups and scale factor language.
Requested order reversed	Name and mark the first quantity before writing either number.
One part used as the whole	Rebuild the complete set and explicitly count every item.
Unlabeled fractional form	Require units or category labels in numerator and denominator.

Small-group routing

If students miss...	Route them to...
Problems 1 or 2	Difference versus factor reteach with equal groups and a number line.
Problems 3 or 5	Ratio-order rehearsal: name the first quantity before writing either number.
Problems 4, 6, or 7	Rebuild complete sets and identify the boundary that represents the whole.
Problem 8 only	Practice finding a missing part before constructing multiple valid ratios.
Problems 7 and 8 are correct	Move to Day 2 error analysis and ratio representation extensions.

Scoring for evidence

Use three evidence marks per problem: 1 point for the correct relationship, 1 point for labels and classification, and 1 point for explanation when requested. Do not treat the total score as mastery by itself; use the error pattern to choose the next instructional move.

Curriculum foundation: Adapted for classroom use from Carnegie Learning Course 1, M2.T1 Skills Practice, pages 43-45, and Lesson 1 Assignment, page 181.